

Team name:	QUANTUMSPEED			

HARDWARE

#	Component	Part name	Autor	Source	Status	Quantity	Unit cost (Euro)	Unit cost (US Dollars)	Total Cost (Euro)	Total Cost (USA Dollars)
1	Microcontroller	Raspberry Pi 5 8GB RAM	Raspberry Pi	https://www.tme.it	Reused	1	151,69	\$175,57	€ 151,69	\$175,57
2	Microcontroller dissipator	Raspberry Pi active cooler	Raspberry Pi	https://www.tme.it	Reused	1	4,72	\$5,46	€ 4,72	\$5,46
3	Memory card	64GB microSD	Sandisk	https://www.tme.it	Reused	1	14,99	\$17,35	€ 14,99	\$17,35
4	Camera	Raspberry Pi Camera Module 3	Raspberry Pi	https://www.tme.it	Reused	2	30,34	\$35,12	€ 60,68	\$70,23
5	Hardware accelerat	Raspberry Pi AI HAT +	Raspberry Pi	https://www.kubii.it	New	1	123,48	\$142,92	€ 123,48	\$142,92
6	Robot chassis	Custom made	QuantumSpeed/BambuLab	https://eu.store.tme.it	New	1	11,50	\$13,31	€ 11,50	\$13,31
7	Microcontroller battery	Ovonic 2S LiPo Battery 5200mAh 7.4V	Ovonic	https://us.tme.it	Reused	1	17,00	\$19,68	€ 17,00	\$19,68
8	Motors battery	Ovonic 2S LiPo Battery 2200mAh 11.1V	Ovonic	https://www.tme.it	New	1	14,00	\$16,20	€ 14,00	\$16,20
9	Drive motor	156.1 Metal gearmotor 20DX44L MM 12V CB	Pololu	https://www.tme.it	New	4	36,92	\$42,73	€ 147,68	\$170,92
10	Motors controller	Dual-Channel H-Bridge Driver Board Module	Akozon	https://www.tme.it	Reused	1	25,86	\$29,93	€ 25,86	\$29,93
11	Motors clamp	20D MM Metal Gearmotor Bracket	Pololu	https://www.tme.it	New	2	9,36	\$10,83	€ 18,72	\$21,67
12	Wheels	Custom made	QuantumSpeed/BambuLab/Reschimica	https://www.tme.it	New	4	5,00	\$5,79	€ 20,00	\$23,15
13	Little servos for arm	180-degree metal gear servo motors	Behogan	https://www.tme.it	Reused	4	1,99	\$2,30	€ 7,96	\$9,21
14	Big servo for arm	270-degree metal gear servo motors 20KG	Diymore	https://www.tme.it	New	1	19,59	\$22,67	€ 19,59	\$22,67
15	Servos controller	PCA9685 Servo driver	AZDelivery	https://www.az-delivery.it	New	1	10,99	\$12,72	€ 10,99	\$12,72
16	Distance sensors	VL53L0X Time-of-Flight	AZDelivery	https://www.az-delivery.it	New	5	3,80	\$4,40	€ 19,00	\$21,99
17	Rotation sensor	Adafruit LSM6DSOX + LIS3MDL	Adafruit	https://www.tme.it	Reused	1	19,95	\$23,09	€ 19,95	\$23,09
18	Microswitches	Micro touch switches 125V 1A	DAOKAI	https://www.tme.it	Reused	8	0,63	\$0,73	€ 5,04	\$5,83
19	Expansion card	PCA9548A Multiplexer I2C	AZDelivery	https://www.az-delivery.it	Reused	1	4,99	\$5,78	€ 4,99	\$5,78
20	Switches	Switches ON - OFF	RS Italia	https://it.rs-online.com	Reused	2	3,61	\$4,18	€ 7,22	\$8,36
21	Servo adapter	I2C Power & Logic Shifter	Melopero	https://www.tme.it	Reused	1	6,05	\$7,00	€ 6,05	\$7,00
22	Power converter	Power converter Step up DC DC	Heemol	https://www.tme.it	New	1	9,50	\$11,00	€ 9,50	\$11,00
23	Power converter	Power converter Step down DC DC	AZDelivery	https://www.az-delivery.it	Reused	1	4,99	\$5,78	€ 4,99	\$5,78
24	LEDs	RGBW Led string	IPIXEL LED	https://www.tme.it	New	1	10,00	\$11,57	€ 10,00	\$11,57
25	Cables	Various cables custom made	QuantumSpeed/OBI Italia	https://www.obitalia.it	New	1	10,00	\$11,57	€ 10,00	\$11,57
26	Connecting parts	screws M1-M2-M3, glue, cable ties, ...	OBI Italia	https://www.obitalia.it	New	1	5,00	\$5,79	€ 10,00	\$5,79

SOFTWARE

#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Euro)	Cost (USA Dollars)
1	Open CV	Library	Image processing library	https://opencv.org/	Open cv.org	FREE	FREE
2	picamera2	Library	Camera control on Raspberry Pi	https://pypi.org/project/picamera2/	Raspberry Pi Ltd	FREE	FREE
3	libcamera	Library	Camera control on Raspberry Pi	https://libcamera.org/	The libcamera Project	FREE	FREE
4	ai_edge_lite	Library	Lightweight AI models for compact devices	https://pypi.org/project/ai-edge-lite/	Google	FREE	FREE
5	tf-lite_runtime	Library	Lightweight AI models for compact devices	https://pypi.org/project/tf-lite-runtime/	Google	FREE	FREE
6	numpy	Library	Fast mathematical calculations on large data arrays	https://numpy.org/	Travis Oliphant	FREE	FREE
7	lgpio	Library	Hardware pin control for electronic components	https://pypi.org/project/lgpio/	Joan (joan2937)	FREE	FREE
8	gpiozero	Library	Hardware pin control for electronic components	https://pypi.org/project/gpiozero/	Ben Nuttall and Dave Jones	FREE	FREE
9	board	Library	Automatic identification of the hardware board model in use	https://pypi.org/project/board/	Adafruit Industries	FREE	FREE
10	adafruit_tca9548a	Library	Management of multiple sensors connected to the same channel	https://github.com/adafruit/Adafruit_TCA9548A/blob/main/adafruit_tca9548a.py	Adafruit Industries	FREE	FREE
11	adafruit_lsm6ds (lsm6ds33)	Library	Accelerometer and gyroscope data reading (movement)	https://github.com/adafruit/Adafruit_LSM6DS33/blob/main/adafruit_lsm6ds.py	Adafruit Industries	FREE	FREE
12	adafruit_lis3mdl	Library	Magnetometer data reading (compass)	https://github.com/adafruit/Adafruit_LIS3MDL/blob/main/adafruit_lis3mdl.py	Adafruit Industries	FREE	FREE
13	time	Library	Management of times, dates, and pauses in the code	https://pypi.org/project/time/	Python Software Foundation	FREE	FREE
14	datetime	Library	Management of times, dates, and pauses in the code	https://pypi.org/project/datetime/	Python Software Foundation	FREE	FREE
15	random	Library	Generation of random numbers or choices	https://pypi.org/project/random/	Python Software Foundation	FREE	FREE
16	signal	Library	Management of clean program interruption	https://pypi.org/project/signal/	Python Software Foundation	FREE	FREE
17	sys	Library	Direct interaction with the operating system	https://docs.python.org/3/library/sys.html	Python Software Foundation	FREE	FREE
18	traceback	Library	Detailed analysis of programming errors	https://pypi.org/project/traceback/	Python Software Foundation	FREE	FREE
19	math	Library	Advanced mathematical formulas and constants	https://pypi.org/project/math/	Python Software Foundation	FREE	FREE
20	pathlib	Library	Simplified management of file and folder paths	https://pypi.org/project/pathlib/	Python Software Foundation	FREE	FREE
21	threading	Library	Simultaneous execution of multiple tasks	https://pypi.org/project/threading/	Python Software Foundation	FREE	FREE
22	Visual Studio Code	Software's Tool	Writing and editing programming code	https://code.visualstudio.com/	Microsoft	FREE	FREE
23	Google Colab	Software's Tool	Running Python code online on Google servers	https://colab.research.google.com/	Google	FREE	FREE
24	Overleaf	Software's Tool	Writing scientific texts and documents in LaTeX	https://it.overleaf.com/	Digital Science	FREE/EDUCATION	FREE/EDUCATION
25	Google Slides	Software's Tool	Creating and editing presentations	https://workspace.google.com/slides/	Google	FREE	FREE
26	Roboflow	Software's Tool	AI dataset management and preparation	https://roboflow.com/	Roboflow, Inc.	FREE	FREE
27	LabelingIMG	Software's Tool	Tool for manual image labeling	https://pypi.org/project/labelingimg/	Tzatalin	FREE	FREE
28	Fusion 360	Software's Tool	Computer-aided 3D design and modeling	https://www.autodesk.com/products/fusion-360/overview	Autodesk	FREE	FREE
29	Bambu studio	Software's Tool	Preparing 3D files for 3D printing	https://bambulab.com/en-us/bambu-studio	Bambu Lab	FREE	FREE
30	Real VNC	Software's Tool	remote access and control software	https://www.realvnc.com/	Andy Harter	FREE	FREE
31	Premiere Pro	Software's Tool	Professional video editing and cutting	https://www.adobe.com/products/premiere.html	Adobe Inc.	€ 1,00	\$1,16
32	After Effect	Software's Tool	Creation of special effects and motion graphics	https://www.adobe.com/products/after-effects.html	Company of Science and Art	€ 1,00	\$1,16
33	Media Encoder	Software's Tool	Fast conversion and rendering of video files	https://www.adobe.com/products/media-encoder.html	Adobe Inc.	€ 1,00	\$1,16
34	Photoshop	Software's Tool	Photo editing and digital image manipulation	https://www.adobe.com/products/photoshop.html	Thomas and John Knoll	€ 1,00	\$1,16

Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Euro EUR	€ 759,60	\$873,37